

ℵ₀ Weekly Problem

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Problem

In how many ways can you tile a $3 \times n$ rectangle by 2×1 dominoes?

Solution

We claim that, for $n \in \mathbb{N}$, the number of ways to tile a $3 \times n$ rectangle using 2×1 dominoes is given by the recurrence $f(n)$ such that:

$$f(n) = \begin{cases} 0, & \text{if } n \text{ is odd,} \\ 1, & \text{if } n = 0, \\ 4f(n-2) - f(n-4) & \text{otherwise.} \end{cases}$$

Proof. First, notice that for odd choices of n , it is impossible to tile the rectangular grid since the placement of a tile reduces the remaining squares in the grid by 2. A successful tiling reduces the remaining squares to 0, but repeated subtraction of 2 grid squares (i.e., dividing n by 2) never results in a remainder of 0. Secondly, there is trivially only 1 way to tile the (nonexistent) grid when $n = 0$: use 0 dominoes.

After establishing an additional base case, we will proceed by strong induction to prove the more general case (i.e., when $n > 0$ and n is even). Namely, we will show that the number of tilings of a $3 \times n$ rectangle using 2×1 dominoes (referred to as simply “dominoes” in the remainder of this document) is given by:

$$f(n) = 3f(n-2) + 2 \sum_{i=4}^n f(n-i),$$

assuming all $f(n-i)$ correctly counts the number of tilings on a grid sized $3 \times (n-i)$, for all integers i where $0 < i \leq n$.

For the purposes of the discussion below, we **zero**-index columns from the **left** when the grid has 3 rows and n columns.

Base Cases:

In addition to the establishment of the base cases above, the following figures are sufficient to show that $f(2) = 3$:

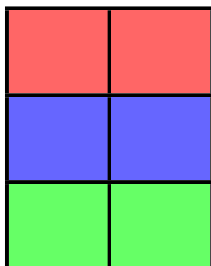


Figure 1

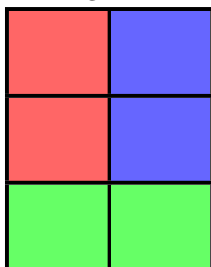


Figure 2

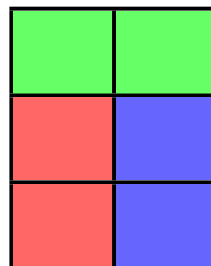


Figure 3

It can also be explicitly shown that $f(4) = 11$.

Inductive Cases:

Our inductive hypothesis is that $f(n) = 3f(n-2) + 2 \sum_{i=4}^n f(n-i)$ correctly counts the number of tilings on a $3 \times n$ grid. We will show that (by assuming $f(n-i)$ correctly counts the number of tilings on a $3 \times (n-i)$ grid for all $i \in \mathbb{N}$ and $n \geq i \geq 4$) there are $3f(n-2) + 2 \sum_{i=4}^n f(n-i)$ unique tilings of a $3 \times n$ grid.

As shown above, $f(2) = 3$. For a $3 \times n$ grid, columns 0 and 1 can be tiled 3 unique ways, leaving an untiled $3 \times (n-2)$ grid. Assuming there are $f(n-2)$

unique ways to tile a $3 \times (n - 2)$ grid, there are $3 \cdot f(n - 2)$ unique ways to tile a $3 \times n$ grid when the first two columns are tiled independently of the other columns. This accounts for the first term of $f(n)$.

We claim that the next term, $2 \sum_{i=4}^n f(n - i)$, counts the tilings where there exists an *offset shape*. An offset shape exists in a tiling whenever the tiling contains one of the 2 following orientations:

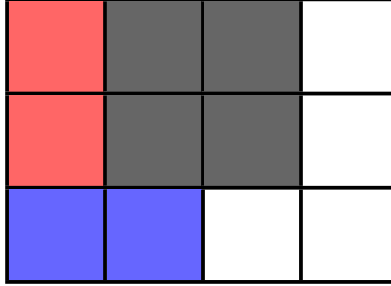


Figure 4

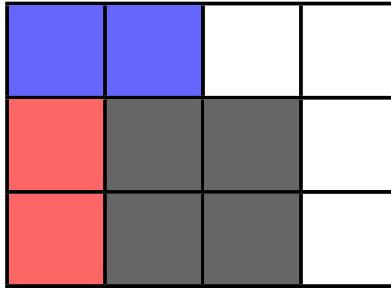


Figure 5

Notice that an offset shape exists whenever exactly two horizontal dominoes in consecutive rows (the gray tiles above) span columns 1 and 2. It can be shown that these dominoes force the subsequent placement of tiles up to (but not including) some column m (for even m and $n > m \geq 4$).

We call the **beginning** of this forced placement of tiles the *initiation* of the offset shape (i.e., the red and blue tiles in the figures herein), and the **end** of this forced placement of dominoes the *resolution* (i.e., the green and yellow tiles in the figures herein).

For example, Figures 6 and 7 (when $n = 6$) show the two possible offset shapes when the initiation of the offset shape has an orientation resembling an “L”. By symmetry, two possible offset shapes exist when the initiation of the offset shape has the other orientation (e.g., Figure 5).

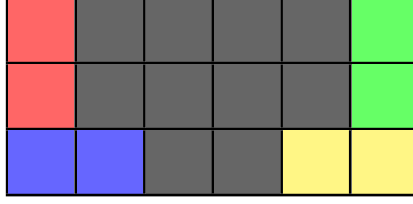


Figure 6



Figure 7

With these preliminaries in place, we can constructively count the tilings that contain an offset shape. Each term in the summation (indexed by i) has a one-to-one correspondence with the case where the offset shape resolves leaving $(n - i)$ columns untiled (e.g., the columns containing white grid squares in Figures 6 and 7) – these yet-to-be tiled columns can be tiled in exactly $f(n - i)$ ways. Since the offset shapes can take two initiation orientations, we multiply the summation by 2 (by symmetry).

Notice that the tilings counted by these two terms are disjoint because the existence of an offset shape requires horizontal dominoes to span columns 1 and 2 (i.e., tilings counted in the second term) whereas the first term counts tilings where horizontal domino(es) span columns 0 and 1. Therefore, the total number of unique tilings of a $3 \times n$ grid is equal to the sum of these two terms, which can be written as:

$$f(n) = 3f(n - 2) + 2 \sum_{i=4}^n f(n - i).$$

Finally, $f(n)$ can be re-written as $f(n) = 4f(n - 2) - f(n - 4)$:

$$\begin{aligned}
f(n) &= 3f(n-2) + 2 \sum_{i=4}^n f(n-i) \\
&= 3f(n-2) + 2f(n-4) + 2 \sum_{i=4}^{n-2} f(n-2-i) \\
&= 3f(n-2) + \left(3f(n-4) + 2 \sum_{i=4}^{n-2} f(n-2-i) \right) - f(n-4) \\
&= 4f(n-2) - f(n-4)
\end{aligned}$$

□